SAECOIN

CONSENSUS SPECIFICATION

Document Version: 1.0
Project: SAECOIN
Ticker: SAC (Provisional)
Blockchain Type: Independent Layer-1
Consensus Mechanism: Proof of Work (PoW)
Transaction Model: UTXO
Document Status: Proposed — Requires Technical Review

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1. PURPOSE

This document defines the proposed consensus protocol for SAECOIN.

The consensus protocol determines:

- How new blocks are created.
- How miners compete to produce blocks.
- How nodes validate blocks.
- How mining difficulty changes.
- How block rewards are distributed.
- How new SAC coins enter circulation.
- How nodes agree on the valid blockchain.
- How invalid blocks and transactions are rejected.

Every compatible SAECOIN node must follow the same consensus rules.

Consensus rules must be deterministic and independently verifiable.

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2. NETWORK PARAMETERS

The following parameters are proposed for SAECOIN.

Parameter| Proposed Value
Blockchain Name| SAECOIN
Ticker| SAC
Blockchain Type| Independent Layer-1
Consensus| Proof of Work
Transaction Model| UTXO
Maximum Supply| 21,000,000 SAC
Initial Block Reward| 50 SAC
Target Block Time| 10 minutes

Halving Interval| 210,000 blocks
Decimal Precision| 8
Premine| 0 SAC

These parameters must be reviewed before implementation.

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3. MONETARY UNIT

SAECOIN will use 8 decimal places.

1 SAC = 100,000,000 smallest units

All monetary calculations must use integer smallest units.

Example:

1 SAC = 100,000,000 units
0.5 SAC = 50,000,000 units
0.01 SAC = 1,000,000 units
0.00000001 SAC = 1 unit

Floating-point arithmetic must never be used for consensus monetary calculations.

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4. PROOF OF WORK

SAECOIN will use Proof of Work.

Miners compete to find a block hash that satisfies the current network difficulty target.

A valid block must satisfy:

Block Hash <= Difficulty Target

The exact hash construction and mining algorithm must be finalized before implementation.

PoW Requirements

- Deterministic block hashing.
- Independent verification by every node.
- Valid difficulty target.
- Valid nonce.
- Protection against invalid block submissions.
- Consistent results across supported platforms.

The final mining algorithm must be selected after security and performance research.

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5. MINING ALGORITHM SELECTION

The mining algorithm has not yet been finalized.

The following requirements will guide selection:

1. CPU mining support.
2. GPU mining support.
3. Reasonable implementation complexity.
4. Strong cryptographic security.
5. Independent verification by full nodes.
6. Open-source implementation availability.
7. Resistance to excessive centralization where practical.
8. Efficient operation on supported hardware.

ASIC resistance is a design objective, not a guarantee.

An algorithm must not be advertised as ASIC-resistant without technical evidence.

The final algorithm selection will be recorded in a separate approved protocol configuration.

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6. BLOCK TIME

Proposed target block time:

10 minutes

The network will attempt to produce one block approximately every 10 minutes.

Actual block intervals will vary because Proof of Work is probabilistic.

The difficulty adjustment mechanism will help maintain the target interval over time.

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7. BLOCK REWARD

Proposed initial block reward:

50 SAC

The miner producing a valid block may receive:

- Block subsidy.
- Eligible transaction fees included in that block.

The block subsidy must follow the halving schedule.

The total coinbase output must not exceed:

Allowed Block Subsidy + Eligible Transaction Fees

Every node must independently verify this condition.

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8. HALVING MECHANISM

Proposed halving interval:

210,000 blocks

The block subsidy will be reduced by half after each halving interval.

Illustrative schedule:

Block Range| Block Reward
0–209,999| 50 SAC
210,000–419,999| 25 SAC
420,000–629,999| 12.5 SAC
630,000–839,999| 6.25 SAC
840,000–1,049,999| 3.125 SAC
1,050,000–1,259,999| 1.5625 SAC

The exact subsidy calculation must use integer arithmetic in the smallest SAC unit.

When further halvings reduce the subsidy below one smallest unit, the subsidy becomes zero.

No floating-point calculations are permitted.

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9. MAXIMUM SUPPLY

Proposed maximum supply:

21,000,000 SAC

The network must enforce this limit.

Supply Rules

- No premine.
- No arbitrary coin creation.
- No unauthorized additional issuance.
- Block rewards must follow the subsidy schedule.
- Transaction fees do not create new coins.
- Invalid coinbase outputs must be rejected.

The total theoretical subsidy must never exceed the maximum supply.

Important Implementation Requirement

The final implementation must calculate the complete issuance schedule before mainnet.

If integer rounding or the final subsidy schedule produces a total different from exactly 21,000,000 SAC, the protocol parameters must be adjusted before launch.

The advertised maximum supply must match the actual consensus-enforced maximum.

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10. DIFFICULTY ADJUSTMENT

Difficulty adjustment maintains the target block interval.

Proposed target:

600 seconds per block

The final difficulty adjustment algorithm is not yet selected.

It must define:

- Adjustment interval or window.
- Expected elapsed time.
- Actual elapsed time.
- Maximum upward adjustment.
- Maximum downward adjustment.
- Timestamp validation.
- Minimum difficulty rules.
- Integer precision and rounding.

The algorithm must resist timestamp manipulation and sudden hashrate changes.

A separate test suite must verify difficulty behavior under:

- Increasing hashrate.
- Decreasing hashrate.
- Sudden miner departure.
- Unstable block intervals.
- Timestamp manipulation attempts.

No arbitrary difficulty formula should be implemented before this section is finalized.

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11. BLOCK VALIDATION

Every full node must independently validate every received block.

Required Validation

1. Block structure is valid.
2. Block version is supported.
3. Previous block reference is valid.
4. Block timestamp follows consensus rules.
5. Proof of Work is valid.
6. Difficulty target is valid.
7. Transactions are valid.
8. Coinbase transaction is valid.
9. Block reward is within permitted limits.
10. No transaction spends an already-spent UTXO.
11. Block size and transaction limits are respected.
12. Merkle root matches the block transactions.

A block that fails any mandatory consensus check must be rejected.

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12. TRANSACTION VALIDATION

Transactions must satisfy the UTXO rules.

Validation Requirements

- Valid transaction format.
- Valid input references.
- Existing unspent outputs.
- Valid signatures.
- Valid spending conditions.
- No duplicate inputs.
- No double spending.
- Valid input and output amounts.
- No monetary overflow.

- Valid transaction fee calculation.
- Valid lock-time conditions.

The sum of transaction outputs must not exceed the sum of transaction inputs for ordinary transactions.

The difference represents the transaction fee, subject to the final fee rules.

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13. CHAIN SELECTION

SAECOIN nodes must use a deterministic chain selection rule.

The proposed rule is:

Select the valid chain with the greatest cumulative Proof-of-Work.

Block height alone must not determine the preferred chain.

Every node must independently verify the blocks and cumulative work.

Chain Reorganization

If a valid alternative chain has greater cumulative work:

1. Identify the common ancestor.
2. Disconnect blocks from the previous active chain.
3. Restore the corresponding UTXO state using undo data.
4. Validate and connect blocks from the new chain.
5. Update the active chain metadata.

Chain reorganization must be handled safely and atomically.

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14. GENESIS BLOCK

The genesis block is the first block of SAECOIN.

It must be defined separately for:

- Development network.
- Testnet.
- Mainnet.

Each network must have a distinct genesis block and network identifier.

Genesis Configuration

The final configuration must include:

- Network ID.
- Genesis timestamp.
- Initial difficulty.
- Initial block version.
- Genesis block contents.
- Genesis block hash.
- Initial consensus parameters.

All nodes on the same network must recognize the same genesis block.

The genesis block must be independently reproducible and verified.

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15. CONFIRMATIONS

A transaction is considered confirmed when it is included in a valid block.

Additional blocks increase the transaction's confirmation depth.

Proposed wallet display policy:

Confirmations| Suggested Status
0| Unconfirmed
1| Confirmed
2–5| More confirmations recommended
6+| Higher-confidence payment

Six confirmations are a suggested payment policy, not absolute finality.

Users and merchants should consider transaction value and reorganization risk.

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16. MINING VALIDATION

Mining software may search for valid solutions, but full nodes must independently verify submitted blocks.

The node must verify:

- Block header.
- Nonce.
- Hash.

- Difficulty target.
- Previous block.
- Transactions.
- Coinbase reward.

A miner must never be trusted solely because it reports a valid solution.

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17. NETWORK CONSISTENCY

All nodes must use the same:

- Consensus rules.
- Block format.
- Transaction format.
- Hash algorithms.
- Difficulty calculation.
- Subsidy calculation.
- Genesis configuration.
- Chain selection rules.

Consensus-critical protocol changes must be documented and introduced through a controlled upgrade process.

Different implementations must produce identical validation results for the same input data.

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18. CONSENSUS ATTACK CONSIDERATIONS

The design and testing process must consider:

51% Attack

An attacker controlling a majority of the network's mining power may be able to reorganize recent blocks and attempt double spending.

Selfish Mining

Miners may attempt to withhold blocks to gain an advantage.

Timestamp Manipulation

Malicious miners may attempt to manipulate timestamps to influence difficulty.

Double Spending

A transaction may be broadcast more than once with conflicting inputs.

Sybil Attacks

An attacker may create many network identities to influence peer connections.

Eclipse Attacks

A node may be isolated from honest peers.

The network must implement appropriate validation, peer management, monitoring, and security measures.

No PoW network can guarantee protection against every attack.

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19. CONSENSUS TESTING

The consensus implementation must include automated tests.

Block Tests

- Valid block.
- Invalid block hash.
- Invalid previous hash.
- Invalid Merkle root.
- Invalid timestamp.
- Invalid difficulty.

Transaction Tests

- Valid transaction.
- Invalid signature.
- Missing UTXO.
- Double spend.
- Invalid output amount.
- Monetary overflow.

Reward Tests

- Initial reward.
- First halving.
- Later halvings.
- Smallest-unit rounding.
- Final subsidy.
- Maximum supply enforcement.

Difficulty Tests

- Stable hashrate.
- Increasing hashrate.
- Decreasing hashrate.
- Sudden hashrate changes.
- Invalid timestamps.

Chain Tests

- Competing valid chains.
- Greater cumulative work.
- Reorganization.
- Invalid alternative chain.

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20. CONSENSUS PARAMETER CHANGE POLICY

Consensus parameters must not be changed casually after mainnet launch.

Any proposed change must include:

1. Technical justification.
2. Security analysis.
3. Compatibility analysis.
4. Testnet validation.
5. Upgrade documentation.
6. Clearly defined activation rules.

Changes affecting consensus require coordinated implementation and community communication.

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21. FINAL PARAMETERS REQUIRING APPROVAL

The following items remain unresolved:

Item| Status
Maximum Supply| Proposed: 21M SAC
Initial Reward| Proposed: 50 SAC
Block Time| Proposed: 10 minutes
Halving Interval| Proposed: 210,000 blocks
Decimal Precision| Proposed: 8
Premine| Proposed: 0
PoW Algorithm| Not Selected
Hash Function| Not Finalized

Signature Algorithm| Not Finalized
Difficulty Formula| Not Finalized
Block Size Limit| Not Finalized
Transaction Size Limit| Not Finalized
Timestamp Rules| Not Finalized
Genesis Configuration| Not Finalized

These unresolved items must be completed before consensus implementation is considered production-ready.

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22. FINAL CONSENSUS PRINCIPLES

1. SAECOIN must remain an independent blockchain.
2. Proof of Work must be independently verifiable.
3. Every node must validate blocks.
4. Maximum supply must be enforced by consensus.
5. Monetary calculations must use integer arithmetic.
6. No unauthorized coin issuance is allowed.
7. The valid chain is selected by cumulative work.
8. Consensus rules must be deterministic.
9. Consensus changes must be controlled and documented.
10. Mainnet must not launch before extensive testing and independent security review.

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23. FINAL NOTES

This document defines the proposed consensus design for SAECOIN.

It is not yet a finalized production protocol.

The mining algorithm, cryptographic primitives, difficulty adjustment formula, exact subsidy implementation, serialization, and network constants must undergo further technical review.

The maximum supply claim must be mathematically verified against the final integer-based issuance schedule before mainnet.

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